

CHAPTER 1

Introduction

1.1. Overview of the book

Attitude Determination and Control Systems are the software, hardware, and algorithms that keep a spacecraft pointed in the desired direction; that is they control the spacecraft's attitude or orientation. Geosynchronous communications satellites need to point their antennas precisely at the Earth to transmit data, voice, television, and data, that produce their revenue stream. A spacecraft docking with the International Space Station needs to keep its docking port aligned with the counterpart on the station. A lunar lander must align its thruster with the desired thrust vector while making sure that its sensors are oriented to receive the data needed for the landing. While often the spacecraft is pointing at one target and rotating slowly with respect to an inertial reference frame, sometimes the control system must rotate the spacecraft quickly.

An attitude control system needs data measuring its orientation with respect to a reference. For a docking spacecraft, such as the Russian Progress, that is the International Space Station. For an Earth-pointing satellite such as a SpaceX Starlink, it is the Earth. For a space telescope, like the Hubble or James Webb, it is an inertial reference, usually the starfield. Often a direct measurement, such as of the attitude with respect to the Earth, is not available with sufficient precision so another measurement, such as the orientation with respect to the star field, is used. The relative orientation with respect to the Earth must be inferred from knowledge of the Earth's rotation with respect to the field of stars. Advanced control systems may be able to identify a target on a planetary or lunar surface and track that target, or in the case of landers, avoid something it does not want to hit.

Once the system has the orientation information, it must compute torques to maintain the orientation or to reorient the spacecraft. This is done with the control actuators such as momentum-exchange devices, thrusters, or magnetic torquers. The disturbances, which perturb the orientation, are generally very small, on the order of micro-N m, but they are applied over long periods, and compensation for their effects is required. They are also a function of the time of year and the orbital position of the spacecraft, further complicating the problem. The actuators, whether thrusters, magnetic torquers, or reaction wheels, tend to be nonlinear and nonideal. The control algorithms couple the measurements to the actuators. While the controller is often just a regulator, which is a controller that maintains the desired orientation, the control algorithms sometimes have to respond to ground commands from operators or the crew onboard.

The dynamics and kinematics of a spacecraft are inherently nonlinear. For small angular rates and small angles from the nominal orientation it is usually possible to

linearize the equations of motion. For an inertially pointing spacecraft, this is a fixed inertial direction. For a planet pointer, this is in a rotating coordinate frame. This lets the designer apply linear control theory.

This book provides an introduction to attitude dynamics, estimation, and control for spacecraft. This book gives you the information needed to solve problems for a wide variety of spacecraft. It sets the design of attitude control systems in the overall spacecraft context. Control systems do not operate in a vacuum and must interact with the other subsystems of the spacecraft, including power, thermal, payload, propulsion, mechanisms, and communications. An attitude control system designer must have a strong background in all of the spacecraft systems since the orientation of the spacecraft affects every subsystem.

The book is organized with the chapters in order of how you design a control system for a spacecraft. Background material such as control theory, actuator and sensor details, and probability and statistics is located in the appendices. The reader can consult the appendices as needed. The book is designed to not require any technical background beyond high-school or freshman-college physics and does not require additional references. MATLAB[®] software is provided that generates all of the plots in the book. The book tries to avoid “black boxes.” The reader is encouraged to read through all of the code that she or he uses.

After the design chapters, the next chapters give detailed designs of real spacecraft based on the author’s design experience. These include a geosynchronous communications satellite, broken into chapters on momentum-bias control and spinning-spacecraft control. Another chapter discusses Sun-nadir pointing control, such as is used on GPS IIR. One chapter covers passive control that is popular for CubeSats and nanosatellites. Another chapter is on lander control. The last chapter is on the James Webb Space Telescope. We provide a “point” solution that is not necessarily the one the James Webb designers used. A chapter on spacecraft simulations is included along with a chapter on spacecraft operations.

The book also includes a chapter on the history of spacecraft. The chapter includes the evolution of spacecraft and satellite design. Most chapters start with a “Space Story” from the author’s experiences in the space business.

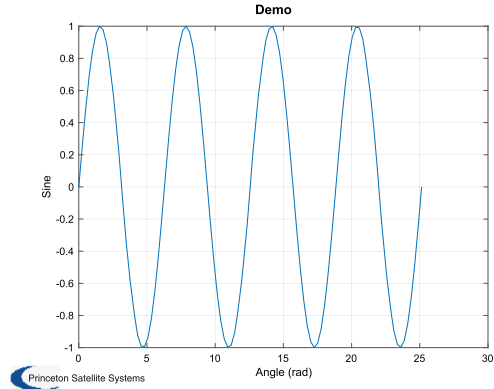
The book includes example code written in MATLAB. Example 1.1 shows a typical example code. In this case, we plot a sine wave. All of the functions used in the example are available for download if they are not built-in MATLAB functions. Since you receive all of the source code you can translate them into your language of choice. For example, going from MATLAB to Python is particularly easy.

You run the script `SetBoothPaths` to set the paths to all of the chapter code and the supporting functions. The script can also run all the scripts and list all of the scripts and functions.

```

1 a = linspace(0,8*pi);
2
3 Plot2D(a, sin(a), 'Angle (rad)', 'Sine', 'Demo');

```



Example 1.1: A sine wave computed in MATLAB. Code examples in this book will be in this format.

For most of the examples, standard MATLAB will suffice. For the Machine Intelligence chapter, the book uses the Deep Learning and Reinforcement Learning toolboxes. These can be purchased as add-ons to basic MATLAB.

The goal of the book is to give you everything you need to be a designer of attitude control systems. Whether you are a high-school student building a CubeSat or an engineer at an aerospace company, this book has something for you. For professors teaching spacecraft control, the book includes three course outlines for different length courses. The author uses this book to teach attitude control at MIT and is happy to answer questions from faculty teaching this very exciting field.

I hope that you enjoy this book!

1.2. Types of spacecraft

Spacecraft come in every imaginable shape and size. Many nations and organizations, including China, Russia, India, the European Space Agency, Japan, and Korea have launched sophisticated spacecraft. Fig. 1.1 shows a selection of NASA spacecraft, both operational and proposed.

GGP Polar is a bias-momentum spacecraft with a despun platform. The spun part has balance masses on lead screws to keep the center-of-mass on the spin-axis. Cassini is a Saturn-orbiting spacecraft. Its control system is called the Cassini Attitude and Articulation Control Subsystem (AACCS) since it controls the articulation of many booms. Cassini is three-axis stabilized using either reaction wheels or thrusters. Cassini also carried the Huygens probe that landed on Titan. Mars Observer was 3-axis stabilized and used reaction wheels and thrusters. Mars Observer was the first NASA spacecraft with reaction wheels. It exploded during its Mars insertion burn. Tracking Data Relay Satellites (TDRS) H, I, and J are NASA geosynchronous communications satellites based on

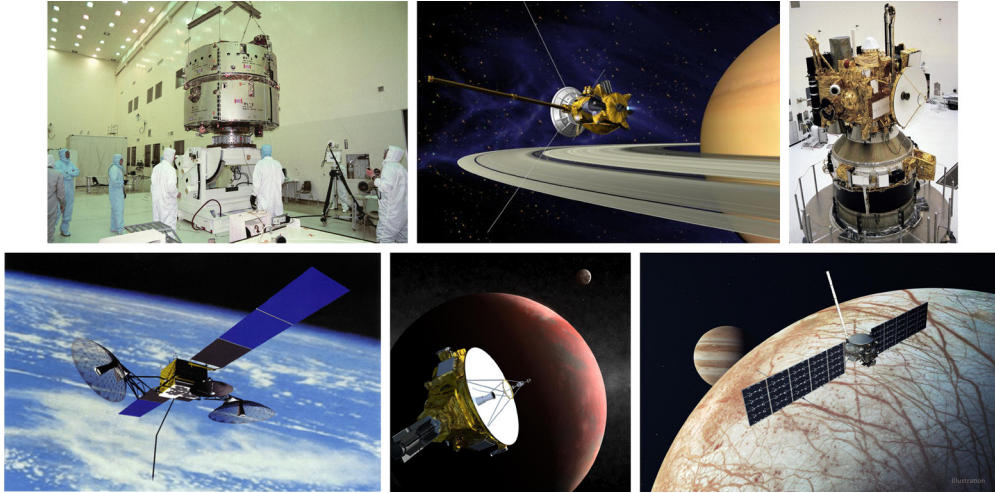


Figure 1.1 NASA spacecraft. From upper left, clockwise. GGS Wind, Cassini, Mars Observer, TDRS H, New Horizons, and Europa Clipper. The author worked on GGS Polar, Mars Observer, and TDRS H, I, J. All images are courtesy of NASA.

the Hughes (now Boeing) HS 601 bus. Bus is a term used for the spacecraft or satellite body where most of the components reside. They use a tilt wheel with two tilt actuators for yaw and roll control. The momentum wheel that is located on the tilt platform is used for pitch control. The solar wings are used for momentum unloading. New Horizons is a NASA interplanetary spacecraft. It flew past Pluto and is still in operation. It is 3-axis stabilized using thrusters. Europa Clipper is a new spacecraft. It has 22 4-N thrusters for control. One of the biggest challenges for Europa Clipper is that it needs to operate in a very high radiation environment.

Fig. 1.2 shows the European Space Agency BepiColombo Mercury Planetary Orbiter at Mercury.

The ACS for BepiColombo is designed with considerable redundancy. It has the following attitude control hardware suite [1,2]:

1. Three star trackers;
2. Two Inertial Measurement Units (IMU);
3. Four fine Sun sensors;
4. Four reaction wheels;
5. Eight 22-N monopropellant hydrazine thrusters for orbit control;
6. Eight 10-N monopropellant hydrazine thrusters for control and momentum unloading.

An interesting feature is the use of a heuristic version of simplex (linear programming) for thruster selection.

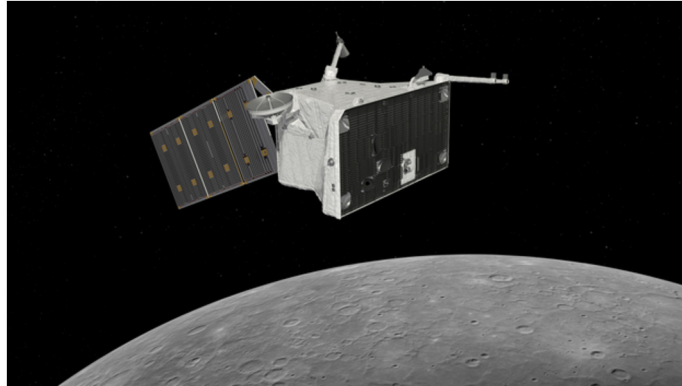


Figure 1.2 European Space Agency BepiColombo Mercury Planetary Orbiter. Copyright: spacecraft: ESA/ATG medialab; Mercury: NASA/Johns Hopkins University Applied Physics Laboratory/Carnegie Institution of Washington.

Fig. 1.3 shows the European Space Agency BepiColombo Mercury Magnetospheric Orbiter at Mercury [3].

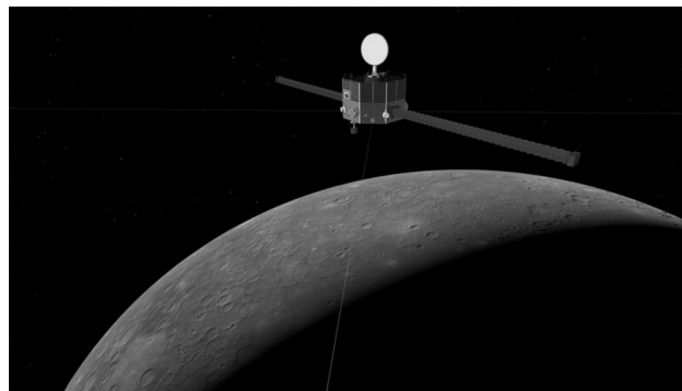


Figure 1.3 European Space Agency BepiColombo Mercury Magnetospheric Orbiter. Copyright: spacecraft: ESA/ATG medialab; Mercury: NASA/Johns Hopkins University Applied Physics Laboratory/Carnegie Institution of Washington.

This spacecraft has a high level of redundancy. This is typical for high-value missions. The Mercury Magnetospheric Orbiter is a spin-stabilized spacecraft.

Table 1.1 gives a list of spacecraft types and their dynamic characteristics.

A gyrostator is a “rigid body” with symmetric rotors.

Table 1.1 Spacecraft examples. CMGs are control-moment gyros. The table is from the simplest attitude control systems design to the most complex design.

Type	Dynamics	Control	Examples
Passive	Rigid body	Gravity gradient, dampers, torquers	GEOS 2.
Spinner	Rigid body	Dampers	Global Geospace Science Wind spacecraft.
Three-axis	Rigid body	Thrusters	Pioneer, Space Shuttle Orbiter, Voyager.
Dual Spin	Gyrostad	Momentum wheel, torquers, thrusters	GE Series 3000–5000, Inmarsat 3, Indostar, BSat, Hughes HS376.
Zero momentum	Gyrostad	Reaction wheels, thrusters, torquers	GPS IIR, DMSP, Tiros, Hubble.
Zero momentum, fast reorientation	Gyrostad	Control-moment gyros, thrusters, torquers	Military Earth-observation satellites.
Zero momentum, with slosh	Gyrostad with pendulum	Reaction wheels, thrusters	Mars Observer.
Multibody gyrostad	Gyrostad with appendages	Pivoted momentum wheel	TDRS H, I, J.
Articulated	Multibody	CMGs	International Space Station (ISS).

TDRS is the NASA Tracking Data Relay Satellite, GPS is the Global Positioning Spacecraft, DMSP is the Defense Meteorological Satellite Program and GEOS is the Geodetic Earth Orbiting Satellite.

Control-moment gyros (CMG)s, are actuators (and not to be confused with gyros, which are sensors). A CMG can be one or two axes. A gyrostad is a type of multibody spacecraft composed of a rigid core and a set of rigid symmetric rotors. A multibody is a spacecraft with points of articulation, much like a terrestrial robot. Joints can be rotational or prismatic (translational). In a gyrostad, rotation does not change the inertia of the spacecraft. In a multibody spacecraft, it does. A gyrostad is a type of multibody spacecraft. More general multibody spacecraft have parts that can translate and rotate with respect to the core spacecraft and the rotations are not necessarily about their center-of-masses. The Space Shuttle Orbiter Remote Manipulator system is an example of a multibody articulated structure. Earth robots are of this type but they differ from spacecraft in that they usually have an Earth-ground reference. Slosh is the movement of a liquid in the spacecraft. Slosh can be due to propellant, heat pipes, or dampers that use a liquid to damp spacecraft angular rates.

Table 1.2 A one-semester course in attitude control systems.

Lecture	Chapter	Description
1	Preliminary Designs	Overview of ACS design. Requirements flow down. Interaction with other subsystems.
2	Control Theory	Introduction to control including loop shaping, Single-input–single-output, and state space.
3	Control Examples	Rate damper, PID, inner and outer loops.
4	Estimation theory background	Filtering, noise.
5	Estimation theory filters	Kalman filters linear, extended, and unscented.
6	Estimation theory examples	Kalman filter examples.
7	Orbit geometry and dynamics	Orbit geometry, orbit representation, orbit dynamics.
8	Orbit maneuvers	Maneuvers impulse, low thrust, close orbits.
9	Kinematics	Quaternions and transformation matrices, ephemeris.
10	Attitude dynamics introduction	Euler's equations.
11	Attitude dynamics advanced	Multibody, slosh, and flex.
12	Mechanisms	Reaction wheels, CMGs, deployment, robot arms.
13	Space environment	Atmosphere, radiation.
14	Budgets	Pointing budgets, disturbance budgets.
15	Actuators	Thrusters, reaction wheels, magnetic torquers, dampers.
16	Propulsion systems	RCS, delta-V, off-pulsing, blowdown.
17	Sensors	Sun sensors, Earth sensors, GPS, magnetometers.
18	Attitude control design	Overview of the process, simple 1-axis example.
19	Stellar attitude estimation	Camera processing, star-data processing centroiding, static solutions Kalman filters.
20	Simulation	Building simulations.
21	Testing	How ACS are tested. Spacecraft test flow.
22	Sun-nadir example	GPS IIR
23	Geosynchronous example	Momentum Bias.
24	Spinning satellite in transfer orbit example	Nutation–damping maneuvering.
25	Zero-momentum spacecraft example	3-axis, star cameras, Sun sensors, reaction wheels, thrusters.
26	Spacecraft operations	Examples of real-time remote operations.
27	Future work in ACS	Machine learning, advanced sensors, and actuators.

1.3. Courses based on this book

1.3.1 A one-semester course

The book fits into a one-semester course with twenty-seven 80-minute lectures. Table 1.2 gives a breakdown of lectures. The book includes MATLAB examples throughout and the students would receive the full source code for all of the scripts. Functions are p-files, that is precompiled functions. Supplementary readings would be technical papers and documents. The book has extensive references but no more than 10 papers would be assigned reading.

This material is used for an MIT course in attitude control. In the author's experience, it is best to focus on the basics in problem sets and projects. In this way, the students develop a base of knowledge that they can use.

1.3.2 A half-semester course

The course can be condensed into a one-semester course or a half-semester course with only 14 lectures. The number of examples is reduced and some lectures are combined. See Table 1.3.

Table 1.3 A half-semester course in attitude control systems.

Lecture	Chapter	Description
1	Preliminary Designs	Overview of ACS design. Requirements flow down. Interaction with other subsystems.
2	Control Theory	Introduction to control including loop shaping, Single-input-single-output, and state space.
3	Estimation theory background	Filtering, noise.
4	Estimation theory filters	Kalman filters linear, extended, and unscented.
5	Orbit geometry and dynamics	Orbit geometry, orbit representation, orbit dynamics.
6	Orbit maneuvers	Maneuvers impulse, low thrust, close orbits.
7	Kinematics	Quaternions and transformation matrices, ephemeris.
8	Attitude dynamics	Euler's equations.
9	Mechanisms	Reaction wheels, CMGs, deployment, robot arms.
10	Space environment	Atmosphere, radiation.
11	Actuators	Thrusters, reaction wheels, magnetic torquers, dampers.
12	Sensors	Sun sensors, Earth sensors, GPS, magnetometers.
13	Attitude control design	Overview of the process, simple 1-axis example.
14	Zero-momentum spacecraft example	3-axis, star cameras, Sun sensors, reaction wheels, thrusters.

1.3.3 An eight-lecture course

The course can be further condensed into an eight-lecture course suitable for an independent activity period or as a short summer course. This is the course the author teaches during MIT's Independent Activity Period. See Table 1.4.

Table 1.4 An eight-lecture course.

Lecture	Chapter	Description
1	Preliminary Designs	Overview of ACS design. Requirements flow down. Interaction with other subsystems.
2	Control Theory	Introduction to control including loop shaping, Single input single output, and state space.
3	Estimation theory filters	Kalman filters linear, extended, and unscented.
4	Orbit geometry and dynamics	Orbit geometry, orbit representation, orbit dynamics.
5	Attitude dynamics and Kinematics	Euler's equations, quaternions, external disturbances.
6	Space environment	Atmosphere, radiation.
7	Actuators and Sensors	Thrusters, reaction wheels, magnetic torquers, dampers, Sun sensors, star cameras, IMUs, GPS.
8	Zero-momentum spacecraft example	3-axis, star cameras, Sun sensors, reaction wheels, thrusters.

References

- [1] ESA, Mercury Planetary Orbiter - spacecraft, <https://sci.esa.int/web/bepicolombo/-/48872-spacecraft>, March 2020.
- [2] L. Szerdahelyi, S. Fugger, P. Espeillac, G. Monroig, T. Pareaud, M. Casasco, The BepiColombo attitude and orbit control system, in: 9th International ESA Conference on Guidance Navigation and Control Systems, 06 2014.
- [3] ESA, Mercury Magnetospheric Orbiter at Mercury, <https://sci.esa.int/web/bepicolombo/-/60079-mercury-magnetospheric-orbiter-at-mercury>, September 2019.